

PROFILE

My work is focused on the modelling and control of complex nonlinear systems that are present in aerospace engineering. This includes robotic systems in manufacturing, metamaterials and autonomous satellite systems. In this area, I have published work in fields such as control theory, multi-agent systems, reinforcement learning and artificial intelligence.

CURRENT ROLE

- **Lecturer (Assistant Professor) in Dynamics & Control** Bristol, UK
School of Engineering, University of the West of England *Sept 2026 - Present*
 - **Role:** I am a lecturer in Dynamics & Control, focusing on applications in mechanical and aerospace engineering. I am interested in the use of metamaterials, optimal control and applied mathematics to solve problems related to autonomous and intelligent systems.
 - **Leadership:** Co-lead of the Space & Aviation Challenge Area for the UK Metamaterials Network.
 - **Collaboration:** I am visiting researcher in the School of Electrical & Electronic Engineering at the University of Sheffield, working in the RAMS Lab and the ATM Lab

GRANTS

- **EPSRC & University of Sheffield:** SURE Project for lattice design of metamaterials (*Principal investigator*)
£5,000 *July 2026*
- **Henry Royce Institute:** Industrial Collaboration Program Grant (*Proposal co-author alongside PI Dr. Simon Pope*)
£88,285 *Mar 2026*
- **EPSRC & Airbus UK:** ICASE Robotics Research Scholarship (*Awarded in recognition of research excellence*)
£35,000 *Sept 2021*
- **RS Online:** Grassroots Student Project Grant on Autonomous Control Systems (*Principal investigator*) - £1,000 *Mar 2021*

EDUCATION

- **Doctorate of Philosophy (Ph.D.), Systems & Control** Sheffield, UK
School of Mechanical, Aerospace & Civil Engineering, University of Sheffield *Aug 2021 - Aug 2025*
 - **Thesis Title:** *Learning-Based Methods for Decision-Making, Planning & Control* working in partnership with Airbus UK.
 - **Assessors:** Dr. Jonathan Aitken (internal) & Prof. Jeremy Watson (external, UCL)
- **Masters of Engineering (M.Eng.), Aerospace Engineering w/ Year in Industry** Sheffield, UK
Department of Aerospace Engineering, University of Sheffield *Sept 2016 - Jul 2021*
 - **Studies:** Graduated with 1st Class (4.0 GPA) degree, studies in avionics, robotics & control theory.
 - **Graduate Thesis:** *Metrology-Assisted Assembly Process to Improve Landing Gear Installation Accuracy*, in partnership with Airbus UK.

EMPLOYMENT HISTORY

Industrial Experience

- **Post-Doctoral Research & Teaching Associate** Sheffield, UK
University of Sheffield *Oct 2024 - Sept 2026*
 - **Role:** Research and teaching associate working on funded projects and being a module leader.
 - **Contribution:** Co-author of a Henry Royce ICP grant with Airbus working on active metamaterials for robotics. Module leader for undergraduate and graduate modules on optimal control, systems engineering and control system design.
 - **Dissemination:** Contributed to publications on reinforcement learning for additive manufacturing, metamaterials and robotics.
- **Research Scientist** Broughton, UK
Airbus Operations Ltd. *Sept 2021 - Jul 2024*
 - **Role:** Research scientist working on developing autonomous solutions for robotics and spacecraft operations.
 - **Contribution:** Assisted on the development and deployment of robotic system for wing assembly and orbital chaser spacecraft.
 - **Dissemination:** Presented work package as part of the *Made Smarter Centre for Connected Factories*.
- **Automation & Robotics Engineer** Weymouth, UK
ASM Assembly Systems Ltd. *Jul 2019 - Jul 2020*
 - **Role:** Worked on developing and deploying robotic systems for semiconductor manufacturing.
 - **Autonomous AGV:** Developed a SLAM-guided AGV for the transportation of large parts and machines.

- **ML Fault Detection System:** Maintained the SQL server used to store all machine test data and wrote the machine learning back-end to predict faults in manufacturing processes.

Durham, UK

Jul 2017 - Sept 2017

Instrument & Test Engineer

CAV Ice Protection Systems

- **Role:** Worked on building, testing and integrating aerospace instruments.
- **“Iron bird” Design:** Managed the deployment of the “Iron bird” digital twin of aircraft control surfaces.
- **Test Hardware Development:** Build the software and hardware for evaluating control surfaces and systems with integrated heating elements.

Internships

Research Intern

Growbotics Ltd.

Didcot, UK

Jul 2024 - Sept 2024

- **Role:** Research intern in the field of autonomous satellites & robotics for sustainable space exploration
- **Collaboration:** Liaised with 3rd party companies and launch providers with producing space-suitable hardware and control interfaces
- **Satellite State Estimation:** Developed low-cost sensor-based state estimators for 6-DOF satellites using MCUs.

Visiting Researcher

ETH Zürich

Zürich, Switzerland

Jul 2022

- **Role:** Invited to be a visiting researcher at ETH Zürich as part of the Robotics Summer School.
- **Topics:** Trajectory optimisation, multi-sensor fusion, SLAM, navigation, full-stack ROS and open source software for deployable robots.

PUBLICATIONS

Journal

- G. Mattera, E. Manoli, **E. Canzini**, and L. Nele, “Integration of reinforcement learning in robotic additive manufacturing control: Advances, challenges, and future perspectives”, *Journal of Manufacturing Processes*, vol. 169, pp. 202–241, Jul. 2026, doi: 10.1016/j.jmapro.2026.04.069.
- **E. Canzini**, D. Miller, D. Tiwari, W. Hutabarat, A. Matthews, and A. Tiwari, “Analysis of sensing modalities for Electrode-Induction Gas atomisation of metal powders”, *Surface and Coatings Technology*, vol. 525, p. 133276, Apr. 2026, doi: 10.1016/j.surfcoat.2026.133276.
- **E. Canzini**, S. Pope, and A. Tiwari, “Geometric visual servo via optimal transport”, *Control Engineering Practice*, vol. 173, p. 106966, Mar. 2026, doi: 10.1016/j.conengprac.2026.106966.
- N. Sanchez-Arriaga, **E. Canzini**, N. Plumb, M. Farnsworth, A. Leyland, and A. Tiwari, “Enhancing thin-film wafer inspection with a multi-sensor array and robot constraint maintenance”, *Nature Scientific Reports*, vol. 15, no. 1, p. 39654, Nov. 2025, doi: 10.1038/s41598-025-23640-5.
- **E. Canzini**, M. Auledas-Noguera, S. Pope, and A. Tiwari, “Decision Making for Multi-Robot Fixture Planning Using Multi-Agent Reinforcement Learning”, *IEEE Transactions on Automation Science & Engineering*, vol. 22, pp. 5578–5589, 2024, doi: 10.1109/TASE.2024.3424677.

Conference

- **E. Canzini**, S. Pope, and A. Tiwari, “Generating Continuous Paths On Learned Constraint Manifolds Using Policy Search”, in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Abu Dhabi, UAE: IEEE, Oct. 2024, pp. 5396–5401. doi: 10.1109/IROS58592.2024.10802531.
- **E. Canzini**, M. Auledas, D. Chasteau, and A. Tiwari, “A Novel Sensing Template Utilising Data Fusion for Large Volume Assembly”, in *14th IFAC Workshop on Intelligent Manufacturing Systems*, Tel Aviv-Yafo, Israel: Elsevier Ltd, 2022, pp. 283–288. doi: 10.1016/j.ifacol.2022.04.207.

Preprint

- M. Raj, B. Brixton, J. Jayakumar, H. Ajraou, **E. Canzini** and R. Ward, “An Extensible ROS2-Gazebo Test-bed Template for Accelerating Robotic Manipulation Research”, Jun. 4, 2026 *Under review at IEEE Robotics & Automation Practice*.

TEACHING EXPERIENCE

Module Teaching (at UoS)

- **Hardware-in-the-Loop & Rapid Control Prototyping (ELE308, 40 students):** 2025-Present
Module leader and lecturer for undergraduate control systems module focusing on deploying real-time controllers to hardware using LabVIEW and MATLAB. Developed the lecture & assessment content.
- **Rapid Control Prototyping (ELE438, 80 students):** 2025-Present
Module leader and lecturer for graduate control systems module focusing on developing optimal control algorithms on embedded hardware. Developed the lecture & assessment content.
- **Industrial Training Programme, Avionics (ACS402, 30 students):** 2021-2023
Module leader and lecturer for graduate avionics module in partnership with Airbus UK. Developed the lecture & assessment content and planned the Airbus interaction.
- **Real-Time Embedded Systems (ACS6127, 80 students):** 2021-2024
Teaching Assistant and Module Leader for graduate control systems module for developing real-time embedded systems. Led labs and tutorial sessions, and delivered lecture content.

Supervision (at UoS)

- **Postgraduate:**
Nathan Espley-Plumb, *Bottom-up Connector-based Assembly for Flexible Assembly Systems*, Ph.D., Feb 2023 - Present. In partnership with Siemens Plc.
- **Graduate:**
Dhiraj Jandu, *Geometric Control and learning-based adaptation for robotic manipulation using force-torque feedback*, M.Eng. thesis, 2026. In partnership with Airbus UK
Joaquin Barawed, *Accurate Feature Detection and Tracking for Relative Pose Estimation in Assembly*, M.Eng. thesis, 2025
Jake O'Neill, *Intelligent Inspection System for Adaptive Robotic Scanning*, M.Sc. thesis, 2025
Elena Manoli, *Towards Industry 5.0: Dynamic Robot Behaviour Manipulation Based on Human Proximity and Movement*, M.Sc. thesis, 2024
- **Undergraduate:**
William Santos, *Development of a Unified Geometric Modelling and Control Framework for the UFACTORY xArms*, B.Eng. dissertation, 2026
Maxwell Bird, *Onboard Relative State Estimation for Proximal Bodies in Orbit*, B.Eng. dissertation, 2025

ACADEMIC CITIZENSHIP

Highlighted Talks

- **“Decision Making for Multi-Robot Fixture Planning Using Multi-Agent Reinforcement Learning”:** May 2025
Presented at the 2025 IEEE International Conference on Robotics & Automation (ICRA), Georgia Institute of Technology
- **“Geometry-Informed Systems for Robotic Manipulation: Methods in Planning & Control”:** May 2025
Presented at the meeting of the Dynamics Research Group, School of Mechanical, Aerospace & Civil Engineering, University of Sheffield
- **“Scaling Robotic Capability In Industry Using Multi-Agent Systems - Applications in Agriculture, Construction & Manufacturing”:** June 2023
Presented at the National Polytechnic Institute of Mexico, Ciudad de México, Mexico

Organised Workshops/Conferences

- **UKMMN+:** Co-chair of the Space & Aviation Challenge Area Research Sandpit 2025
- **University of Sheffield:** Co-chair of the ACSE Research Symposium 2022

Professional Activities

- **HEA:** Fellow of the Higher Education Academy (FHEA) 2024 - Present
- **IEEE:** Member of the Institute of Electrical & Electronic Engineers (MIEEE) 2020 - Present
- **RAeS:** Member of the Royal Aeronautical Society (MRAeS) 2026 - Present